Supporting Information for "Minimum-Seismicity Injection Design Is Convex, and the Value of Design Is Set by Two Numbers"

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Contents: Text S1 (θ-law, three-arc optimum), Text S2 (synthetic set-up and verification), Text S3 (pre-registered field tests), Text S4 (simulator design and coupled case), Table S1, Figures S1–S2.

Text S1. Three-arc optimum and the θ-law

Time is normalised by T_{inj} and the volume to one. The stress obeys $\theta\dot{s} = u - s$, $s(0) = 0$, with $u \in [0, u_{\max}]$, $\int_0^1 u\,\mathrm{d}t = 1$; the tail decays freely. Per unit background rate, $J = \int_0^\infty (e^{\Pi s/s_c} - 1)\mathrm{d}t$ with $s_c = 1 - e^{-1/\theta}$. The Hamiltonian is linear in u; on a singular arc $\partial H/\partial u \equiv 0$ implies $\dot{\lambda} = 0$ and hence a constant stress s^* with $u = s^*$. The optimum is: charging at $u_{\max}$ for $t_r = \theta \ln(u_{\max}/(u_{\max} - s^*))$, plateau, and a maximum-rate burst before shut-in (for $u_{\max} = 10$, $\theta \gtrsim 0.1$ a coasting arc can precede it). Constant rate gives $J_c = (e^\Pi - 1) - \theta e^\Pi \,\mathrm{Ein}(\Pi) + \theta[\mathrm{Ei}(\Pi) - \gamma - \ln\Pi]$ for $\theta \ll 1$, with $\mathrm{Ein}(x) = \int_0^x (1 - e^{-y})/y\,\mathrm{d}y$. To first order, $G = 1 - J^*/J_c = \theta\kappa_2(\Pi, u_{\max}) + O(\theta^2)$ with $\kappa = e^\Pi[\Pi m + R - \mathrm{Ein}(\Pi)]/(e^\Pi - 1)$, $m = (u_{\max} - 1)\ln(u_{\max}/(u_{\max} - 1))$, R the start-up layer term. A direct discretised optimal-control solution agrees with the closed form to $\sim 10^{-5}$; the first-order term is accurate to +5–8% up to $\theta = 0.1$ (script `research/exp/h1f_theta_law.py`).

Text S2. Synthetic set-up and verification

3-D point-source diffusion with exact erfc step responses; $k = 10^{-14}$ m^2, $\mu = 10^{-3}$ Pa s, diffusivity 0.1 m^2/s, friction 0.6, $A\bar{\sigma} = 0.05$ MPa, $t_a = 100$ yr, patch at 300 m, 8640 m^3 in 10 days, rate bound three times the mean, 100 control intervals, counting to five injection times. Optimisation: SLSQP with the analytic adjoint gradient (softmax weights). Verification: ODE vs integral form to 10^{-5}; 2000 midpoint convexity tests (Figure S1a); gradient vs finite differences; multistart; KKT residual; dose invariance to 0.1%. Excess events (shut-in / total): constant 16.36 / 38.83, step-up 12.21 / 40.49, taper 20.81 / 40.16, optimum 14.14 / 37.01.

Text S3. Pre-registered field tests

Pipeline: kernel (multi-exponential, rate → wellhead pressure), Dieterich with threshold P^* and diffusion time τ_d, counts above M_c. Plans were written before any fit (`research/CYCLE3_PLAN.md`, `CYCLE6_PLAN.md`). These are internal plan files, not deposited with an external registry, so their dates are not independently time-stamped; they are archived unchanged with the code.

Test	Criterion	2018	2020
E1 kernel hold-out	RMSE $\leq$10% of range	PASS (6.1%)	WEAK (11.4%)
E2 Dieterich fit	corr $\geq$0.98, tail $\pm$30%	WEAK (tail +50%)	PASS (0.998, +8%)
E3 temporal hold-out	end $\pm$30%	FAIL (+29% shut-in)	FAIL (degenerate P^*)
E4 gain optimum vs constant	2020: $\leq$3% predicted	1.6%	**9.1% (refuted)**
E8 θ-scan peak	in [0.05, 1]	0.26	**0.26 (confirmed)**
E9 transfer of 2018 parameters	corr $\geq$0.95	—	PASS (0.985)
E6 design from hold-out	keeps gain	PASS (70%)	FAIL (2.9$\times$)

2020 values: $V = 2884$ m^3, shut-in 387.3 h, 6121 events ($M_c = -1.45$, $b = 1.39$), $P^* = 72.9$ bar, $\tau_d = 10.0$ h, $\Pi = 3.43$; θ-scan gains 2.4 / 7.6 / 14.0 / 11.5 / 5.8% at $\theta = 0.026$ / 0.078 / 0.26 / 0.77 / 2.58. With pauses imposed: constant 3816, optimum 2747 events.

Text S4. Simulator design and coupled case

Table S1: designed vs constant rate on a heterogeneous 41×41 case.

	Constant rate	Designed
Expected events, design model	127.9	100.2
Expected events, simulated	143.8	117.8 (-18.0%)
Model error	17.8% (first pass)	0.15% (after one correction)

Coupled case (`examples/FAULT_INJECTION.DATA`): $30 \times 20 \times 5$ cells of $50 \times 50 \times 10$ m at 1975 m, fault with 100 faces (MULTFLT 0.05) 150 m from a water injector (800 sm^3/d, 280 bar cap, shut-in at 500 d); S_v gradient 0.24 bar/m, $S_{hmin}/S_v = 0.65$, $S_{Hmax}/S_v = 1.10$ at 60°, stress path 0.5, $\mu = 0.6$, $t_a = 100$ yr, $r_0 = 5$ km^{-2}yr^{-1}, $b = 1$; two-way factor 20. Maximum ΔCFS 14.6 (two-way) vs 11.2 bar; 483 sampled events, largest M_w 2.27 (Figure S2). A rate-and-state trace ($a = 0.010$, $b = 0.015$, $D_c = 1$ mm) produced no dynamic rupture ($h^* = 920$ m exceeds the 1-km trace).

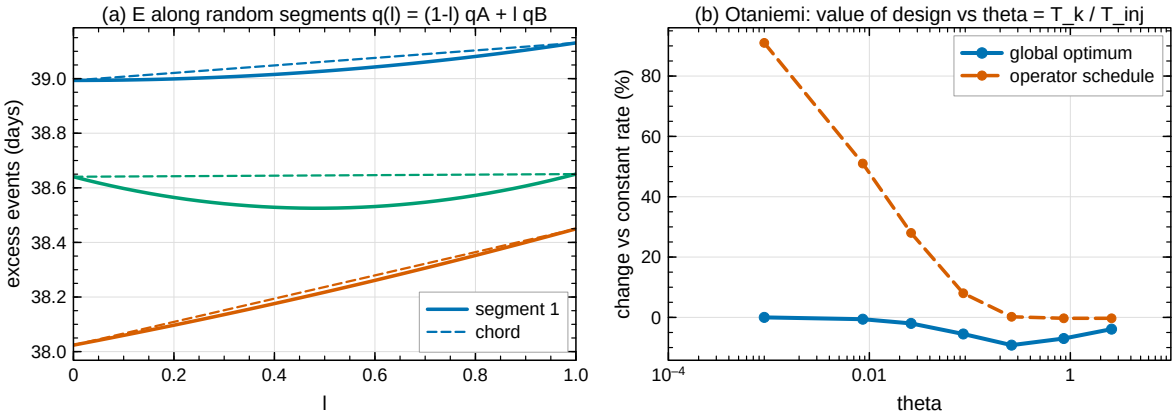


Figure 1: *

Figure S1. (a) Expected events along segments between random feasible schedules lie on or below the chord. (b) Otaniemi 2018: change relative to constant rate of the optimum and of the operator schedule versus θ.

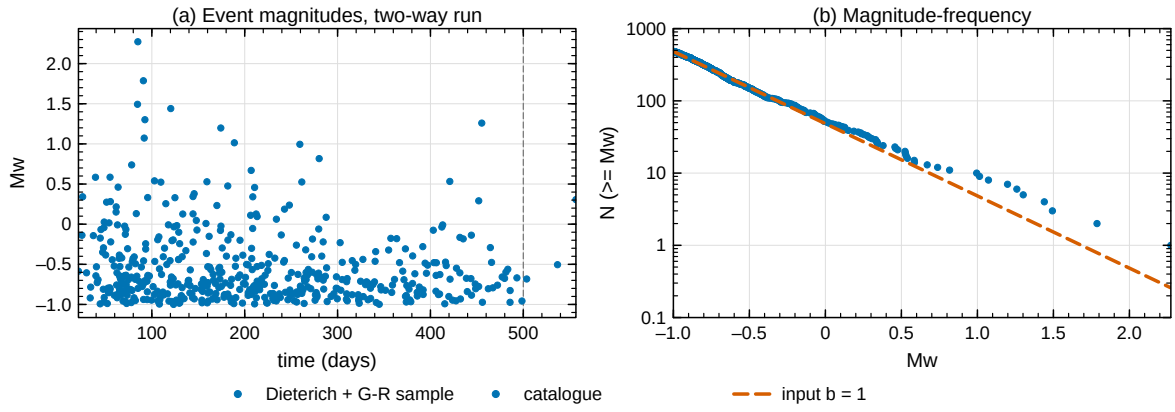


Figure 2: *

Figure S2. Stochastic catalogue of the two-way run: magnitudes versus time and magnitude–frequency distribution (input $b = 1$).